

Ques 3:

$$E_x = E_y = E_z = (n_x + \frac{1}{2})\hbar\omega$$

(1) $E_N = \frac{(N+3)\hbar\omega}{2}$

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$$N = n_x + n_y + n_z$$

ex: $N=0$, $E_0 = \frac{3}{2}\hbar\omega$: $n_x = n_y = n_z = 0$

$N=1$: $E_1 = \frac{4}{2}\hbar\omega$

$n_x, n_y, n_z : \{1, 2, 3, 4, \dots\}$

$$N=1 = n_x + n_y + n_z$$

$$1 = n_x + n_y + n_z$$

3 case: $N=1$

$$N=1 : 3 \text{ cases}$$

$$3N$$

- (i) $n_x=1$, $n_y=0=n_z$
- (ii) $n_y=1$, $n_x=0=n_z$
- (iii) $n_z=1$, $n_x=0=n_y$

$N=2$: $2 = n_x + n_y + n_z$

1	1	0
1	0	1
0	1	1
2	0	0
0	2	0
0	0	2

$N=2 : 6$

$N=3$: 3

n_x	n_y	n_z
1	1	1
2	1	0
0	2	1
0	1	2
1	0	2
1	2	0
3	0	0
0	3	0
0	0	3

$9 = 3N$

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$$g = 3N$$

(4) $\hat{L}_z = \hbar (x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x}) \rightarrow \hat{L} = \hbar \hat{x} \hat{y}$

Polar: $\begin{cases} x = r \cos(\phi) \\ y = r \sin(\phi) \end{cases}$

$\hat{r} = i \hat{r} \frac{\partial}{\partial x}$

Chain Rule: $f = f(g(x))$
 $\frac{d}{dx} f(g(x)) = \left(\frac{df}{dz} \right) \frac{dz}{dx}$

$g(x) = z$

$\frac{df(g(x))}{dx} = \frac{dz}{dx} \frac{df(z)}{dz}$

$\frac{d}{dx} = \frac{dz}{dx} \frac{d}{dz}$

$(x, y) \rightarrow (r, \phi)$
 (z_1, z_2)
 $r \quad \phi$

$\frac{d}{dx} = \sum_i \frac{dz_i}{dx} \frac{d}{dz_i}$

$\frac{d}{dx} = \frac{dz_1}{dx} \frac{d}{dz_1} + \frac{dz_2}{dx} \frac{d}{dz_2}$

$z_1 = r, z_2 = \phi$

$\begin{cases} \frac{\partial}{\partial x} = \left(\frac{\partial r}{\partial x} \right) \frac{\partial}{\partial r} + \left(\frac{\partial \phi}{\partial x} \right) \frac{\partial}{\partial \phi} \\ \frac{\partial}{\partial y} = \left(\frac{\partial r}{\partial y} \right) \frac{\partial}{\partial r} + \left(\frac{\partial \phi}{\partial y} \right) \frac{\partial}{\partial \phi} \end{cases}$

$\hat{L}_z = -i \hbar (x \partial_y - y \partial_x)$

$\frac{d}{d\phi} = \frac{\partial x}{\partial \phi} \frac{\partial}{\partial x} + \frac{\partial y}{\partial \phi} \frac{\partial}{\partial y}$

$x = r \cos(\phi)$

$y = r \sin(\phi)$

$\frac{\partial x}{\partial \phi} = -r \sin(\phi) = -y$

$\frac{\partial y}{\partial \phi} = r \cos(\phi) = x$

$\frac{\partial}{\partial \phi} = -y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y}$

$$\underline{L_z} = -i\hbar \frac{\partial}{\partial \phi} = -i\hbar \left(-y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y} \right) \\ = -i\hbar (x \partial_y - y \partial_x)$$

chp 7:

(1) Three quantum numbers

$$n: \hat{H}, E_n$$

$$l: \vec{L}^2$$

$$m: \hat{L}_z$$

$$\bullet l = n - 1$$

$$E_n \leftrightarrow n$$

$$E_n = \left(\frac{N+3}{2} \right) \hbar \omega, \quad N=1, \dots$$

$$\bullet N = n_x + n_y + n_z$$

$$\text{Hydrogen: } E_n = -C \frac{1}{n^2}$$

$$n: l, \text{ or } m$$

(a) $n = l + 1$

(b) $m: -l, \dots, l$

Using sums: $2l+1$ states for all n

$$\sum_{l=0}^{n-1} (2l+1) = n^2$$

$$m = -l \dots -1 \ 0 \ 1 \dots l$$

$$\underbrace{\quad \quad \quad}_{l} \quad \underbrace{\quad \quad \quad}_{l}$$

$$l + 1 + l = 2l + 1$$

For every n :

$$\sum_{l=0}^{n-1} l = \frac{(n-1)n}{2}$$

Radial wave fns:

$$n \quad l \quad R_{n,l}$$

$$1 \quad 0 \quad R_{1,0}(r) = 2a_0^{-3/2} e^{-r/a_0}$$

$\vdots$

$\vdots$

$$(7.3.3) \rightarrow (2.3.5) \quad R(\rho) = \rho^L e^{-r/\rho} f(\rho)$$

↓ assume

$$f(\rho) = \sum c_j \rho^j$$

$$R(\rho) = \rho^L e^{-r/\rho} \sum c_j \rho^j$$

$$* a_{n\ell m} = R_{n\ell}(r) Y_{\ell m}(\theta, \phi)$$

$$dx dy dz \rightarrow \underline{dr} d\phi d\theta \underline{r^2 \sin \theta}$$

$$|n, \ell, m\rangle$$

$$\bullet \int dr r^2 |R_{n\ell}|^2$$

$$\underline{n=3} \quad \underline{R_{3,2}}: * C_{j+1} = \frac{2\gamma(1+j+\ell)-2}{(j+1)(j+2\ell+2)} C_j \quad * (7.3.10)$$

$$j=0 \quad C_1 = \frac{2\gamma(1+\ell)-2}{(2\ell+2)} \quad \gamma = \frac{1}{n}$$

$$* C_1 = \frac{(1+\ell)-1}{n} C_0 \quad \rho = r/a_0$$

$$R_{1,0} = C_0 e^{-r/a_0}$$

$$\{ \begin{matrix} n \geq 1 \\ \ell = 0 \end{matrix} \quad R(r) = \left(\frac{r}{a_0}\right)^\ell e^{-\frac{r}{na_0}} f\left(\frac{r}{a_0}\right)$$

$$R(\rho) = \rho^\ell e^{-\rho} f(\rho) \\ \downarrow \\ \sum_{i=0}^{\infty} C_i \rho^i \quad *$$

$$= C_0 \left(\frac{r}{a_0}\right)^0 + C_1 \left(\frac{r}{a_0}\right)^1 + \dots$$

$$* R_{2,0} = \left(\frac{r}{a_0}\right)^0 e^{-r/a_0} C_0 + \left(\frac{r}{a_0}\right)^1 e^{-r/a_0} C_1$$

$$R_{1,0} = e^{-r/a_0} C_0$$

$$R_{2,0} = \left(e^{-r/a_0}\right) \left[C_0 \left(\frac{r}{a_0}\right)^0 + C_1 \left(\frac{r}{a_0}\right)^1 \right]$$

$$* C_1 = \left(\frac{(1+\ell)-2}{n(2\ell+2)}\right) C_0 \quad (n=2, \ell=0)$$

$$C_1 = -\frac{1}{2} C_0$$

$$R_{3,\ell} = \left(\right) \left(C_0 \left(\frac{r}{a_0}\right)^0 + C_1 \left(\frac{r}{a_0}\right)^1 + C_2 \left(\frac{r}{a_0}\right)^2 \right)$$

$$\bullet R_{3,2} = \left(\frac{r}{a_0}\right)^2 e^{-r/a_0} \left(\right)$$

$$C_2 = \frac{(1+\ell)-1}{n(2\ell+2)} C_1 = \left(\right) (-1/6) C_0$$

$$c_2 = \left(\frac{2}{3 \cdot 2} \right) c_1 = \left(\frac{1}{3} \right) \left(-\frac{1}{2} c_0 \right)$$

$$l = 2$$

$$c_2 = \left(\frac{2}{3 \cdot 2} \right) c_1 = \frac{1}{3} \left(-\frac{1}{2} c_0 \right)$$

$$R_{3,2} = \left(\frac{r}{a_0} \right)^2 e^{-r/3a_0} c_0 \left(1 + -\frac{1}{2} \left(\frac{r}{a_0} \right) - \frac{1}{6} \left(\frac{r}{a_0} \right)^2 \right)$$

ah another error

I'll write this up

